INCH-POUND

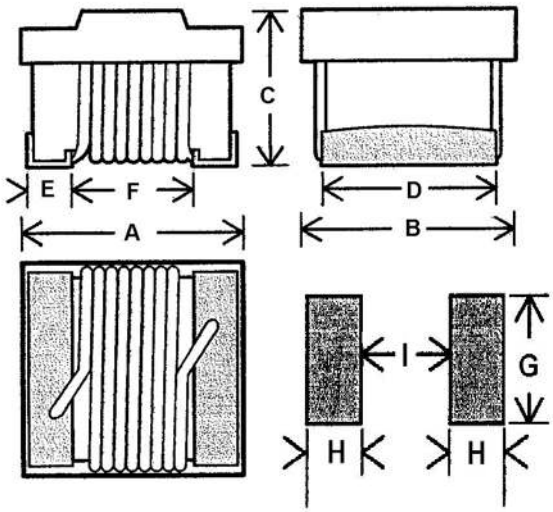
MIL-PRF-83446/37
6 June 2003

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, OPEN CONSTRUCTION, SURFACE MOUNT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-83446.



LAND PATTERN

	INCHES	MILLIMETERS
A	0.090 Max.	2.29 Max.
B	0.068 Max.	1.73 Max.
C	0.060 Max.	1.52 Max.
D	0.050 (Ref. Only)	1.27 (Ref. Only)
E	0.018 (Ref. Only)	0.44 (Ref. Only)
F	0.040 (Ref. Only)	1.03(Ref. Only)
G	0.070	1.78
H	0.040	1.02
I	0.030	0.76

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Electrical characteristics (initial).

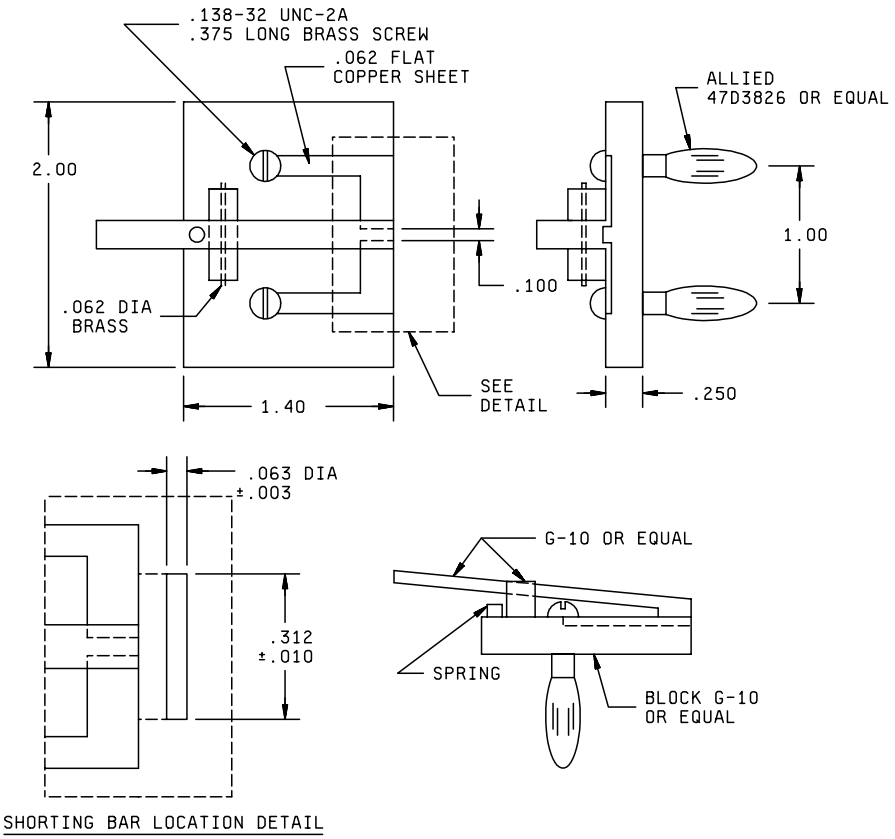
Dash Number	Inductance nH ±10%	Inductance Test Frequency MHz	Q (min)	Q Test Frequency MHz	Self Resonant Frequency MHz (min)	DC Resistance Ohms (max)	Current Rating mA (max)
-01	2.8	250	80	1500	7900	0.06	800
-02	3	250	65	1500	7900	0.06	800
-03	3.3	250	50	1500	6000	0.08	600
-04	5.6	250	65	1000	5500	0.08	600
-05	6.8	250	50	1000	5500	0.11	600
-06	7.5	250	50	1000	4500	0.14	600
-07	8.2	250	50	1000	4700	0.12	600
-08	10	250	60	500	4200	0.10	600
-09	12	250	50	500	4000	0.15	600
-10	15	250	50	500	3400	0.17	600
-11	18	250	50	500	3300	0.20	600
-12	22	250	55	500	2600	0.22	500
-13	24	250	50	500	2000	0.22	500
-14	27	250	55	500	2500	0.25	500
-15	33	250	60	500	2050	0.27	500
-16	36	250	55	500	1700	0.27	500
-17	39	250	60	500	2000	0.29	500
-18	43	200	60	500	1650	0.34	500
-19	47	200	60	500	1650	0.31	500
-20	56	200	60	500	1550	0.34	500
-21	68	200	60	500	1450	0.38	500
-22	82	150	65	500	1300	0.42	400
-23	91	150	65	500	1200	0.48	400
-24	100	150	65	500	1200	0.46	400
-25	110	150	50	250	1000	0.48	400
-26	120	150	50	250	1100	0.51	400
-27	150	100	50	250	920	0.56	400
-28	180	100	50	250	870	0.64	400
-29	220	100	50	250	850	0.70	400
-30	240	100	44	250	690	1.0	350
-31	270	100	48	250	650	1.0	350
-32	330	100	48	250	600	1.4	310
-33	390	100	48	250	560	1.5	290
-34	470	50	33	100	375	1.7	220
-35	560	25	23	50	340	1.9	210
-36	620	25	23	50	220	2.2	210
-37	680	25	23	50	200	2.2	190
-38	750	25	23	50	200	2.3	180
-39	820	25	23	50	200	2.35	180

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable percent variation from the initial measurement			
Qualification Inspection:	Inductance	Q	Self-resonant Frequency	DC resistance
Group II	±5	-10	-8	±3 (+.001 ohm)
Group IV	±5	-10	-10	±3 (+.001 ohm)
Group V	±5	-10	-----	-----
Quality Conformance Inspection:				
Group C				
Subgroup II	±5	-10	-10	±2 (+.001 ohm)
Subgroup IV	±5	-10	-8	±3 (+.001 ohm)

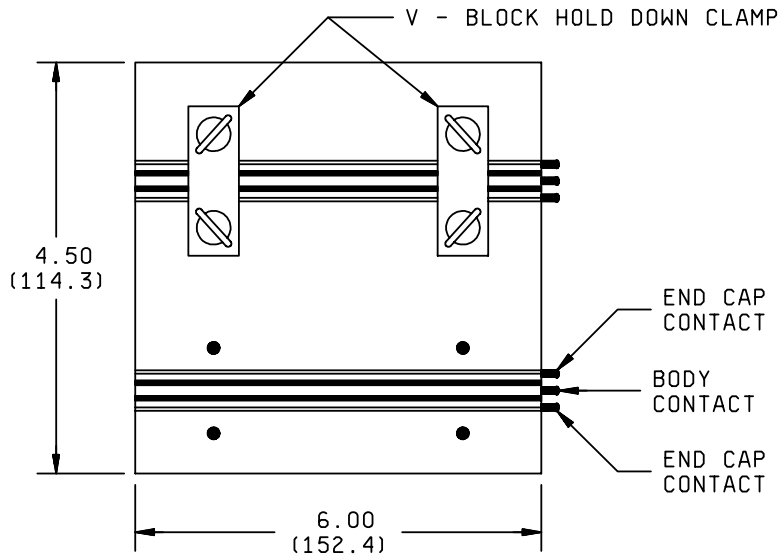
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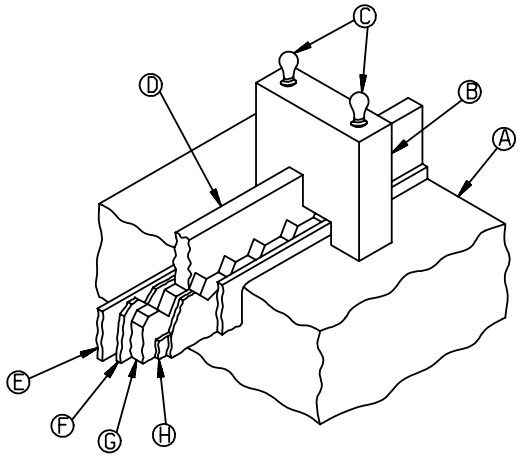


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Shorting bar material is brass.
4. The total fixture and residual inductance is approximately .049μH (for references only).

FIGURE 2. General test fixture, (TF-397).

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ENLARGED CUT-AWAY OF CLAMP

Legend	
LTR	Description
A	G-10 base
B	Upper V-block hold-down clamp
C	Clamp screws
D	Upper V-block (body contact)
E	End cap contact retaining wall
F	End cap contact V-block (in contact with E)
G	Lower V-block (body contact)
H	G-10 insulators (separating G from F/E)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Test fixture for insulation resistance, dielectric withstanding voltage and barometric pressure.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Bond strength: Force 2 pounds.

Weight: 0.19 gram maximum.

Operating temperature range: -40°C to +125°C.

Temperature rise (at +90°C): +15°C.

Maximum operating temperature: +125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 500 Vrms for 40 seconds.

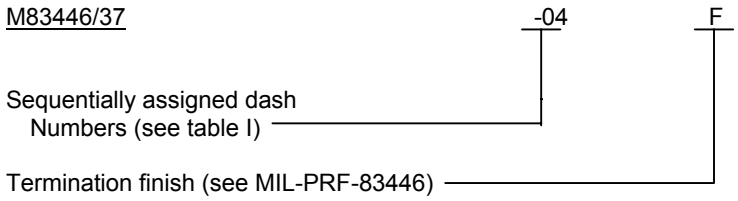
Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 200 Vrms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Test fixture: Tests shall be performed using test fixtures shown on figure 2 and 3, or equivalent.

Part or Identifying Number (PIN): The PIN shall be in the following form:



Custodians:
Army - CR
Navy - EC
Air Force - 11

Review activities:
Army - MI
Navy - AS, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5950-1149)

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